

Anjitha Mekkayil Anand

Senior .NET Engineer | Agentic AI · RAG · LLM Evaluation | C# · Azure · MCP

[LinkedIn](#) · [GitHub](#) · [anjitha-mekkayil-anand.vercel.app](#)

PROFILE

Senior engineer with 12 years of production experience, building agentic AI systems in .NET. Hands-on across the AI-native stack: RAG pipelines with grounded cited answers (Anamnesis — measured baseline: hit-rate@5 0.95, MRR 0.90, faithfulness 1.00), multi-agent orchestration with MCP tool servers (Mathesis, Microsoft Agents League hackathon), multi-provider LLM routing with automatic failover, and LLM-as-judge evaluation harnesses. Applies GenAI daily in production workflows — Claude Code, agentic review pipelines, and human-in-the-loop design — on a large enterprise energy platform. Open to AI-native engineering roles where production discipline and measured quality matter.

TECHNICAL SKILLS

Languages & Frameworks	C#, .NET 10, ASP.NET Core, Blazor, Web API, Entity Framework Core, KnockoutJS, JavaScript, HTML, CSS, LINQ, ADO.NET
AI & Agentic Systems	RAG pipelines (embeddings, vector search, grounded citation), LLM evaluation (retrieval metrics, LLM-as-judge faithfulness), multi-provider LLM routing & fallback, MCP (Model Context Protocol), Claude API, OpenAI API, agentic workflow design, tool surface design, human-in-the-loop systems, AI-augmented development (Claude Code, GitHub Copilot)
Cloud & Infrastructure	Microsoft Azure — IoT Hub, AI Search, Functions, Cosmos DB, SQL Server, Web Apps, Azure AD, Event Hubs, Azure DevOps (CI/CD) · Docker · Observability — OpenTelemetry, Azure Monitor, Application Insights
Databases & Tools	MS SQL Server (query optimization, index design, performance tuning), SQLite, SSMS, Visual Studio, Git
Patterns & Practices	Clean Architecture, CQRS, Repository, Decorator, Modular Monolith, Multi-tenant Architecture, Microservices, OAuth 2.0 / OpenID Connect, Agile / Scrum

PROFESSIONAL EXPERIENCE

Senior Software Developer — EG India, Mangaluru

Aug 2023 – Present

Enterprise energy management platform · .NET / Blazor / SQL Server · Nordic utility clients

- Delivered features, architectural improvements, and production fixes across five modules of a large, multi-tenant, multi-language .NET/Blazor/SQL Server modular monolith serving Nordic utility operators — billing workflows, metering data pipelines, and market protocol integrations — across **279 merged pull requests**
- Built **EG ClarityHub** — a full-stack AI-integrated CRM platform — in .NET 10 and Blazor for Talent Accelerator 2026, one of 26 across EG's Industrials & Trade division. Presented live to Exec VP, CTO, and SVP panel; panel response: *"We'd like this in production today."* **Runner-Up** across development, product, support, and commercial evaluation
- Reviewed **812 pull requests** as the cross-team review path, with formal accountability for work delegated to junior engineers
- Root-cause analysis across frontend, backend, and database layers; SQL Server and EF Core query optimization and index design, including an N+1 pattern reduced from one call per record to two calls total, regardless of record count
- Refactored a 7,000-line front-end module where adding a single status meant hand-editing 17 hardcoded arrays — replaced them with a three-layer resolution pipeline, surfacing 12 pre-existing bugs including a silently incorrect status-group mapping, and cutting the two largest files by ~75%
- Versioned REST API development for external client integrations
- Applied AI-augmented development practices using Claude Code, MCP tool integration, and agentic workflows; established patterns for AI-assisted root-cause analysis and introduced the approach to the team

C# · .NET 10 · Blazor · KnockoutJS · SQL Server · Entity Framework Core · Azure

Senior Software Engineer — [Zowork, Bengaluru \(Remote\)](#)

Aug 2022 – Jul 2023 · 11 mos

- Built the real-time device-to-client event pipeline on Apache Kafka and Azure Event Hubs across distributed systems in .NET Core
- Implemented OAuth 2.0 and OpenID Connect authentication and authorisation with IdentityServer4 across a microservices platform, securing inter-service communication
- Designed and maintained RESTful backend services in .NET Core and C# across a distributed microservices architecture — API design, performance, and data integrity for enterprise clients
- Mentored junior developers through code review across the platform's backend services

C# · .NET Core · SQL · Azure Web App · Azure SQL DB · OAuth · Microservices · Kafka · Dapper · MediatR

.NET & Cloud Developer — [WeMakeloT, Calicut, Kerala](#)

May 2019 – Aug 2022 · 3 yrs 3 mos

- Designed and deployed an Azure IoT monitoring platform across 2 enterprise clients — IoT Hub, Functions, Cosmos DB, Azure AD, AppInsights and Power BI dashboards — covering water infrastructure and industrial IoT across rural and commercial deployments
- Migrated a legacy ASP.NET application to .NET Core for a global financial services client, modernising the stack and improving maintainability
- Built .NET Core web APIs and websites hosted on Azure, with CI/CD through Azure DevOps
- Built ESP32 device firmware and web server for remote monitoring and control, bridging device-level work with the cloud platform

C# · .NET Core · ASP.NET · Entity Framework · Azure IoT · Cosmos DB · Azure Functions · Azure AD · SQL · JavaScript

Senior Developer & Team Lead — [OHO Software Innovations, Cochin](#)

Apr 2017 – Apr 2019 · 2 yrs

- Converted a live billing product to India's GST regime at launch — learned the new tax rules directly from the legislation, mapped them onto the existing billing model, and shipped to 10+ enterprise clients
- Led migration of a legacy FoxPro billing system and its DBF databases to .NET C# and SQL Server, evolving it into an integrated billing and reporting platform
- Led a team of ~6 engineers through the full SDLC — requirements through to production delivery
- Delivered SSRS reports, complex stored procedures, and T-SQL for an ERP platform

C# · WinForms · ASP.NET · ADO.NET · SQL · SSRS · JavaScript · jQuery · IIS

.NET & SQL Server Developer — [OHO Software Innovations, Cochin](#)

Jan 2016 – Apr 2017 · 1 yr 3 mos

- Built and maintained intranet and ERP applications in ASP.NET and SQL Server
- Developed complex stored procedures, user-defined functions, and SSRS reports for data analysis

C# · ASP.NET · ADO.NET · SQL · SSRS · JavaScript · jQuery · IIS

Project Trainee Engineer — [Gadgeon Smart Systems Pvt. Ltd., Thrikkakara](#)

Aug 2014 – Jun 2015 · 10 mos

- M.Tech project internship in real-time computing: compared response times across Xenomai, ARM Cortex-A8 and the PRU subsystem on AM335x, then built a real-time data collection library using Programmable Real-time Units — achieving deterministic timing without an RTOS
- Cross-compiled the Linux kernel for AM335x and BeagleBone Black; built a plotting, logging and alert system by embedding Python in C
- Work contributed to two published research papers (ACM WCI'15 and IJERT)

Embedded C · Linux · PRUSS · AM335x

Software Engineer — [Infosys, Mangaluru](#)

Sep 2006 – Feb 2008 · 1 yr 6 mos

- Mainframe development for enterprise clients — COBOL, DB2, JCL batch processing, CICS online screens

COBOL · DB2 · JCL · CICS

PERSONAL PROJECTS

Anamnesis — RAG Pipeline with Evaluation Harness (Jul 2026 · github.com/anjitha-mekkayil-anand/anamnesis)

Retrieval-augmented generation over my LinkedIn posts and Noesis Letters newsletter archive · Grounded cited answers · Multi-provider routing with failover · Measured quality baseline · .NET 10

- Built a RAG service end-to-end: paragraph-aware chunking with overlap, OpenAI embeddings stored in SQLite, exact cosine top-k retrieval, and grounded answers with inline citations
- Provider router with Polly resilience policies — Claude primary, retry with exponential backoff and per-attempt timeout, automatic failover to OpenAI
- Evaluation harness with a 20-question golden set: retrieval hit-rate@k and MRR, LLM-as-judge answer faithfulness, per-stage latency — results logged per run as a quality trend
- First baseline: hit-rate@5 0.95, MRR 0.90, faithfulness 1.00 — the grounding guard declines rather than inventing when retrieval misses

C# · .NET 10 · RAG · Embeddings · SQLite · Claude API · OpenAI API · Polly · xUnit

Mathesis — Multi-Agent Enterprise Learning System (Jun 2026 · Microsoft Agents League hackathon · github.com/anjitha-mekkayil-anand/Mathesis)

Four specialised reasoning agents · Custom MCP tool server · Azure AI Search grounded retrieval · Human-in-the-loop approval gates · .NET 10

- Designed and built a multi-agent enterprise learning system end-to-end: orchestrator, skills advisor, content curator, and assessment designer agents
- Custom MCP tool server exposing learning domain tools to Claude reasoning agents via Model Context Protocol
- Azure AI Search grounded retrieval with semantic ranking; human-in-the-loop approval gates for all agent recommendations
- Built solo in .NET 10 with full unit test coverage; submitted Jun 2026

C# · .NET 10 · MCP · Claude API · Azure AI Search · SQLite · xUnit

Pronoia — Predictive Maintenance AI System (2026 · github.com/anjitha-mekkayil-anand/Pronoia)

Multi-phase .NET 10 system: sensor simulator → change detector → MCP server → Claude reasoning agent → Blazor human approval queue

- **Pronoia.Detection:** SensorEvaluator + ChangeDetector — three states (Normal / Drift / Breach); drift on running average catches the pre-breach pattern predictively
- **Pronoia.Mcp:** MCP server with guardrail-by-tool-surface design — agent can only read and flag for review, never act directly
- Closes a 20-year arc: BTech factory automation (2006) → ACM-published PRUSS system (2015) → AI reasoning agent (2026)

C# · .NET 10 · SQLite · MCP · Claude API · Blazor · xUnit

CERTIFICATIONS

Microsoft Azure AI Engineer Associate (AI-102)	Jun 2025 – Jun 2027
Anthropic Academy — Claude Code · MCP · Agent Skills	2026
GitHub Copilot Certification (GH-300)	Dec 2025 – Dec 2027
Microsoft Applied Skills: Accelerate AI-assisted development by using GitHub Copilot	Oct 2025
Professional Scrum Master I (PSM I)	Jun 2025 · no expiry
Microsoft Azure Developer Associate	Feb 2021 – Feb 2027

EDUCATION

Master of Technology (M.Tech.) — Embedded Systems	2015
National Institute of Electronics & Information Technology (NIELIT), Calicut	
<i>Full-time programme, 2013–2015 — preceded by career break for family responsibilities and relocation.</i>	
Bachelor of Technology (B.Tech.) — Applied Electronics & Instrumentation	2006
Government Engineering College, Kozhikode	

ADDITIONAL INFORMATION

Location Mangaluru, India · Open to remote and relocation for the right opportunity

Work authorisation Indian national · Eligible to apply for relevant work visas
Languages English (Professional) · Malayalam (Native)

PUBLICATIONS

Using PRUSS for Real-Time Applications on Beaglebone Black

WCI '15 — Proceedings of the Third International Symposium on Women in Computing and Informatics · Cited 11×
(Google Scholar)

<https://dl.acm.org/doi/10.1145/2791405.2791408>

Real-time Application in Linux using PRUSS

International Journal of Engineering Research & Technology · Cited 2× (Google Scholar)

<https://www.ijert.org/research/real-time-application-in-linux-using-pruss-IJERTV4IS050971.pdf>